

Weekly Report – The Hybrid Model

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This Week

- **Two U-Nets compared:** Akeret et al.'s tf_unet (ours) vs. the U-Net of Mesarcik et al.
- **Hybrid model:** presented for the first time — a U-Net with 9 changes.
- **Which changes matter:** tested by removing parts and re-training.

Key Results (LOFAR, max F1)

- **tf_unet:** 0.548
- **Mesarcik et al. U-Net:** 0.588 (+0.039)
- **Our hybrid:** **0.660 — highest of all methods**
- **Hybrid with 3 parts removed:** 0.659 — no real loss

➤ Two different U-Nets

Akeret et al. (2017) — tf_unet

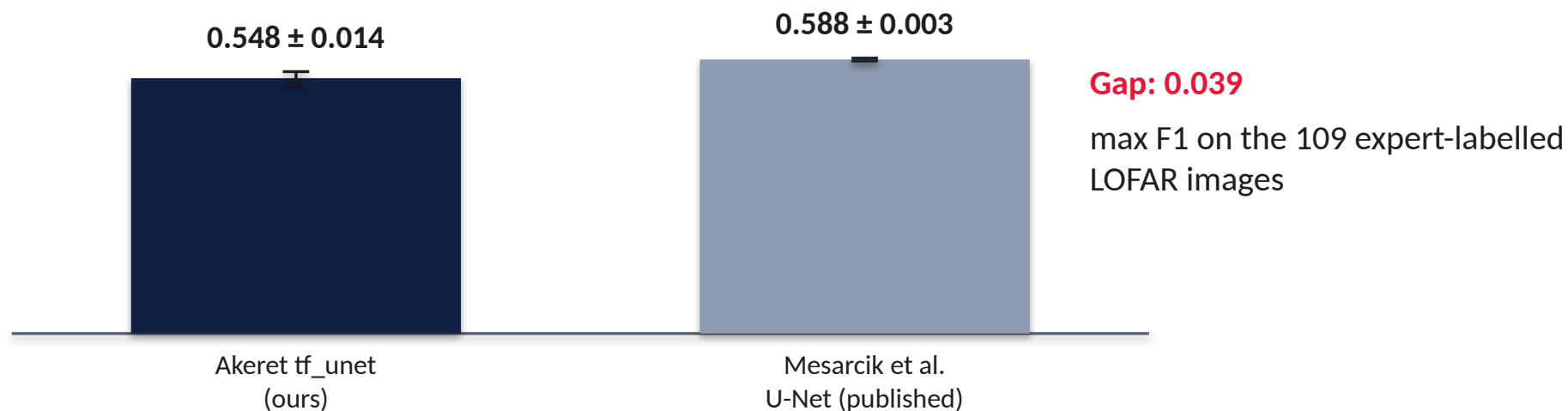
The original U-Net code for RFI detection, written in TensorFlow.

This is the U-Net we ran.

Mesarcik et al. (2022) — their own U-Net

Rebuilt from scratch by the authors of the LOFAR dataset paper.

Same name, but a different network (see next slide).



➤ What is different between them

	Akeret et al. tf_unet (ours)	Mesarcik et al. U-Net
Normalisation layers	None	BatchNorm after every layer
Shrinking the image	Max-pooling (3 levels)	Stride-2 convolutions (5 levels)
Image edges	Trimmed (valid padding)	Kept (same padding)
Network size	465,986 parameters	1,179,121 parameters
Training input	Whole 512×512 image, 4 per batch	32×32 patches, 1024 per batch
Input scaling	Fixed linear range	Clipped, then log-scaled
Output	2 scores, ReLU, softmax	1 score, sigmoid

So the 0.039 gap compares two different networks — not the same U-Net trained twice.

➤ Our Hybrid model: 9 changes to the U-Net

Group 1 — RFI-specific parts (3)

1. **Strip convolutions** — look along long thin lines
2. **Residual shortcuts** — a bypass around each block
3. **Channel attention (ECA)** — turns useful features up

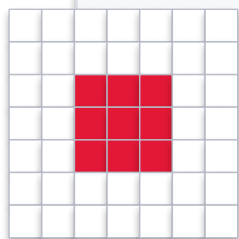
Result: max F1 **0.660 ± 0.004** on LOFAR, with 593,842 parameters.

Group 2 — training & design changes (6)

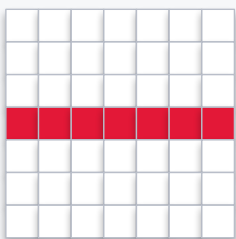
4. **GroupNorm** — keeps internal values steady
5. **No ReLU on the output** — avoids the dead-network trap
6. **Dice loss** — rewards the right RFI shape
7. **Same padding** — predicts every pixel, edges too
8. **No class weighting** — doesn't over-push towards RFI
9. **New size & training** — 4 levels, thinner, PyTorch

➤ The three RFI-specific parts

Strip convolutions



normal filter



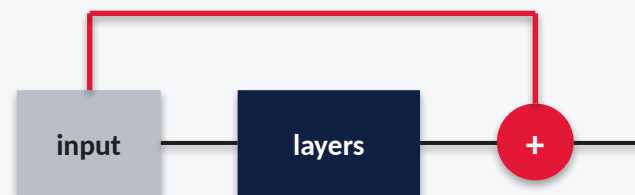
strip filter

Look along long thin lines (7, 11 or 21 pixels) instead of small squares.

Idea: RFI often appears as streaks.

Residual shortcuts

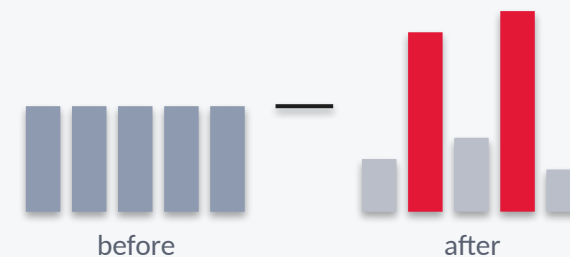
shortcut



A bypass road: each block passes its input straight through and only learns a small correction.

Idea: faint details are not lost in deep layers.

Channel attention (ECA)



Learns which features matter and turns them up or down.

Idea: let faint-RFI features stand out.

➤ Full hybrid vs. partial hybrid

Full hybrid

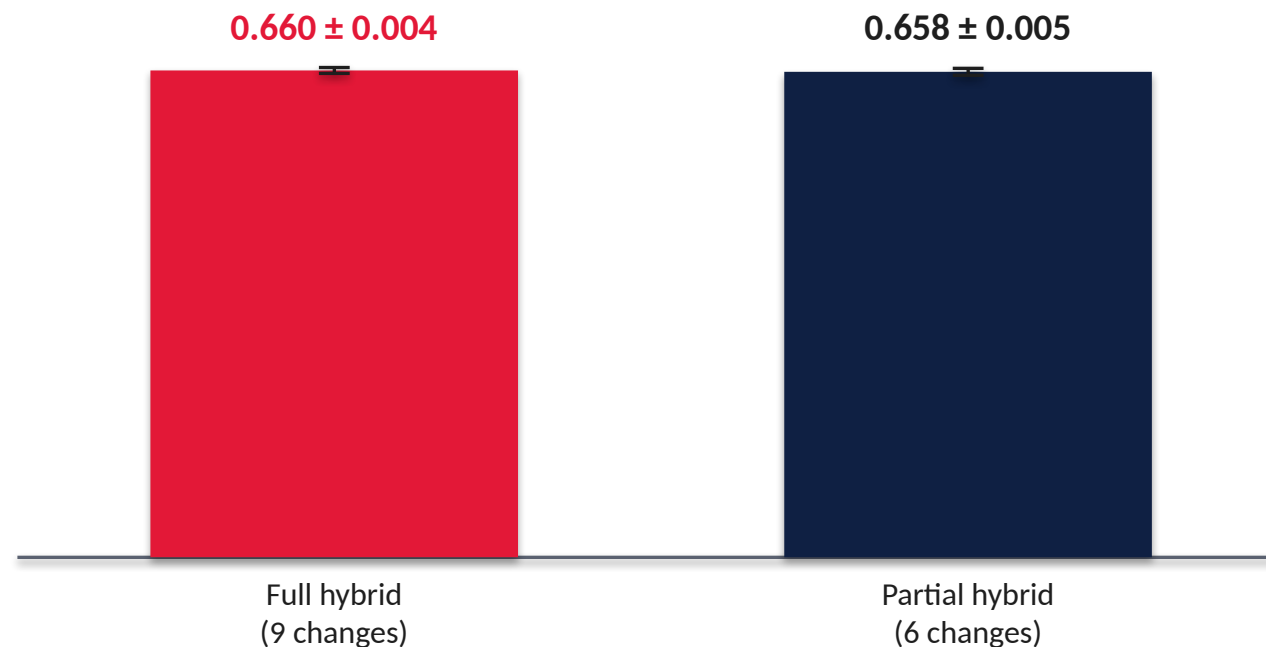
All 9 changes

593,842 parameters

Partial hybrid

Group 1 removed: no strips, no shortcuts, no attention (6 changes left)

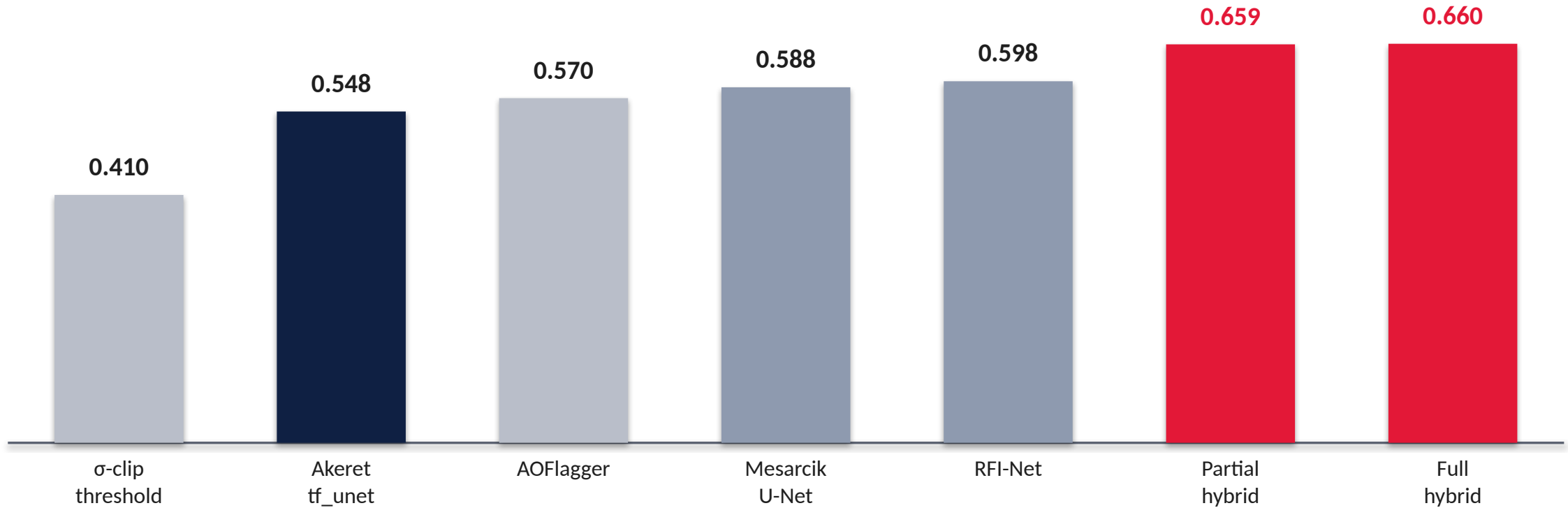
486,418 parameters (18% smaller)



Difference: 0.002 — not significant (3 runs each, $p = 0.64$).

The three RFI-specific parts add 18% more parameters but no accuracy.

➤ Where everything stands on LOFAR



max F1 on the 109 expert-labelled LOFAR images. Mesarcik et al. U-Net and RFI-Net values as published by Mesarcik et al. (2022).

➤ To Do Next Week

- **Find what drives the gain:** test the 6 Group-2 changes one at a time, starting with the Dice loss.
- **Repeat runs** where results so far come from a single run.
- **Project report:** continue from the full first draft.